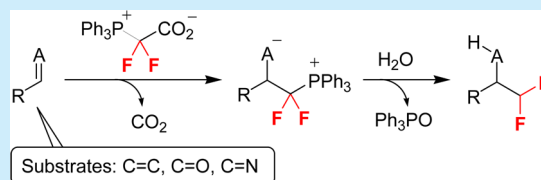


Difluoromethylene Phosphobetaine as an Equivalent of Difluoromethyl Carbanion

Vitalij V. Levin,[†] Alexey L. Trifonov,[‡] Artem A. Zemtsov,[†] Marina I. Struchkova,[†] Dmitry E. Arkhipov,^{§,⊥} and Alexander D. Dilman^{*,†}[†]N. D. Zelinsky Institute of Organic Chemistry, Leninsky prosp. 47, 119991 Moscow, Russian Federation[‡]Moscow Chemical Lyceum, Tamozhennyj proezd, 4, 111033 Moscow, Russian Federation[§]A. N. Nesmeyanov Institute of Organoelement Compounds, Vavilov str. 28, 119991 Moscow, Russian Federation[⊥]N. I. Pirogov Russian National Research Medical University, Ostrovityanova str. 1, 117997 Moscow, Russian Federation

S Supporting Information

ABSTRACT: A method for nucleophilic difluoromethylation of reactive Michael acceptors, aldehydes, and azomethines is described. The reaction is performed using the readily available and air-stable reagent difluoromethylene phosphobetaine. The process involves interaction of an electrophilic substrate with in situ generated difluorinated phosphonium ylide followed by hydrolysis of the carbon–phosphorus bond under mild conditions.

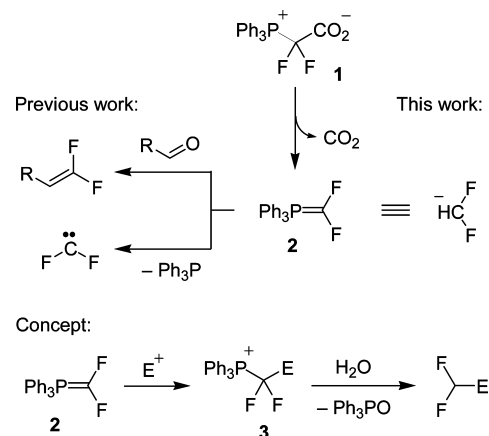


Organofluorine compounds play important roles in medicinal chemistry and related fields, as evidenced by the great number of fluorine-containing drugs and agrochemicals.¹ While the trifluoromethyl group has for a long time been the most frequently used and studied fragment,^{1,2} today partially fluorinated groups attract significant attention. For example, the difluoromethyl substituent (CHF₂) exhibits unique pharmacoforic properties that make it capable of serving as a lipophilic hydrogen bond donor, thus being bioisosteric to the hydroxyl group.³

Nucleophilic fluoroalkylation has emerged as a reliable methodology for the direct introduction of fluorinated groups into organic molecules.^{2a–c,4} Although it is well developed for trifluoromethylation,^{2a–c} difluoromethylation is more challenging to perform.^{4–7} To this end, several reagents for the introduction of CHF₂-carbanion were described, but limited substrate scope, high reagent cost, and availability issues constitute major limitations of existing methods.⁸ In this work, we describe a robust difluoromethylation protocol involving inexpensive, shelf-stable, and easy-to-handle reagent.

Recently, Xiao reported preparation of difluoromethylene phosphobetaine (**1**) from triphenylphosphine and potassium bromodifluoroacetate⁹ (Scheme 1). While a stable crystalline compound at room temperature, upon mild heating, betaine **1** undergoes decarboxylation to generate transient phosphonium ylide **2**.^{10,11} The latter species has not been identified,¹¹ but it can react with aldehydes affording olefination product^{9,10a,11,12} or it can expel phosphine to generate difluorocarbene.^{10b,c} Herein we demonstrate that ylide **2** can serve as an equivalent of difluoromethyl carbanion.¹³ Our concept is based on using ylide **2** as a nucleophile followed by hydrolytic cleavage of the C–P bond in intermediate phosphonium salt **3**, thereby

Scheme 1. Use of Difluorinated Phosphonium Ylide



affording a product of nucleophilic difluoromethylation (Scheme 1, bottom equation).

Given our interest in performing fluoroalkylation of electron-deficient alkenes,¹⁴ we selected arylidene derivatives of Meldrum's acid (**4**) owing to high electrophilicity of the double bond.^{14b,15} Thus, the interaction of compound **4a** with betaine **1** was performed in dimethylformamide at 60 °C (Scheme 2). The formation of phosphonium intermediate **5a** can be proved by ¹⁹F NMR spectroscopy. Zwitterionic species **5a** was isolated in individual state and fully characterized, and its structure was confirmed by single-crystal X-ray analysis (Figure 1). A solution of generated phosphonium salt **5a** was treated with water (ca. 10 equiv) followed by heating to effect

Received: November 6, 2014

Scheme 2. Difluoromethylation of Substrate 4a

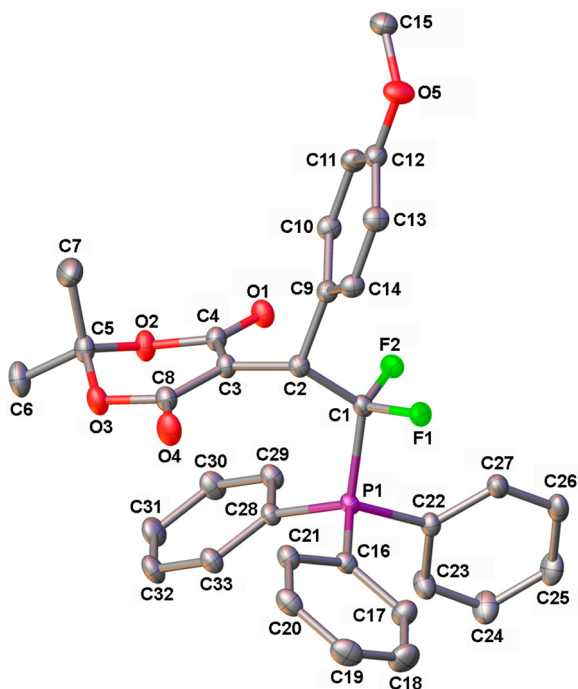
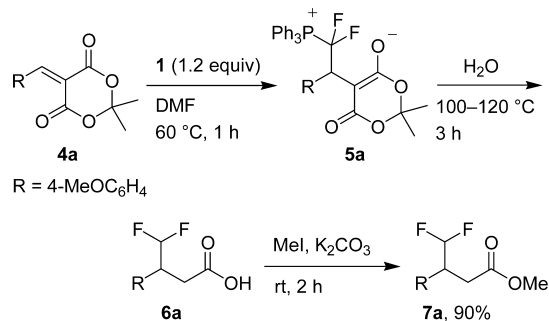
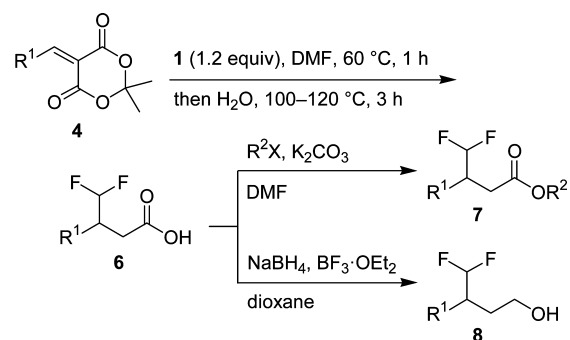


Figure 1. X-ray structure of zwitterion 5a. Hydrogen atoms are omitted for clarity.

hydrolysis of the C–P bond along with transformation of Meldrum's acid moiety into carboxylic acid. In fact, protodephosphorylation proceeded at room temperature (vide infra), but heating was required for decarboxylation. It proved to be difficult to purify carboxylic acid 6a by chromatography owing to significant tailing on silica gel, and crude acid was converted to methyl ester 7a, which was isolated in 90% yield based on starting 4a.

A series of arylidene substrates 4 reacted with betaine 1 with subsequent protodephosphorylation (Table 1). The intermediate acids 6 were either esterified (using MeI or BnBr) to afford esters 7 or reduced with a sodium borohydride/boron trifluoride combination to afford alcohols 8. Reactions worked well with substrates derived from aromatic aldehydes bearing electron-donating or electron-withdrawing substituents. Only for compound 4e having an *o*-methoxy group was product 7e formed in reduced yield of 54% (entry 5). Rewardingly, alkylidenes 4j,k obtained from enolizable aliphatic aldehydes gave good yields of products 7j,k (entries 10 and 11). Apparently, the successful addition to enolizable substrates is associated with relatively low basicity of phosphonium ylide 2. However, the reaction of a substrate derived from Meldrum's

Table 1. Reactions of Aryl- and Alkylidene Meldrum's Acids 4



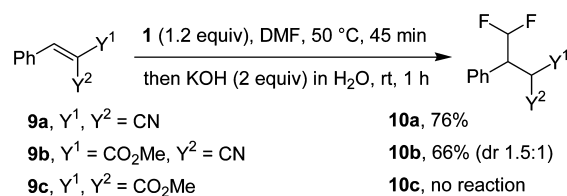
no.	substrate	product	yield, % ^a
1			7b 75
2			8c 78
3			7d 88
4			8d 84
5			7e 54
6 ^b			7f 93
7 ^b			7g 76
8			7h 78
9 ^c			7i 75
10 ^d			7j 75
11			7k 71

^aIsolated yield based on substrate 4. ^bAqueous KOH at 60 °C was used for the protodephosphorylation step. ^cMeCN was used as solvent in the reaction with 1. ^d2 equiv of 1 was used.

acid and phenyl methoxycarbonyl ketone afforded no difluoromethylation product.

The reactivity of other Michael acceptors was briefly evaluated (Scheme 3). Benzylidene derivatives of malononitrile and cyanoacetic ester 9a,b gave difluoromethylated products 10a,b in reasonable yields. At the same time, benzylidene malonate 9c was unreactive because of its low electrophilicity.¹⁶ It should be pointed out that for substrates 9a,b, proto-

Scheme 3. Difluoromethylation of Substrates 9



dephosphorylation of intermediate phosphonium salts was performed at room temperature using 2 equiv of aqueous alkali.

It was interesting to investigate difluoromethylation of aldehydes and azomethines since resulting difluorinated alcohols and amines are useful building blocks. *N*-Tosylimine, iminium ion, and difluoroboryl complex derived from *N*-benzoylhydrazone¹⁷ gave corresponding products **12a–c** in good yields (Table 2, entries 1–3). The progress of the

Table 2. Difluoromethylation of C=O and C=N Bonds

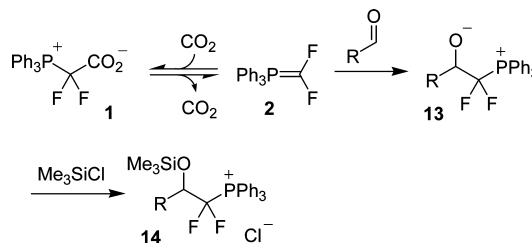
no.	substrate	T, °C	product	yield, % ^a
1 ^{b,c}	11a	55	12a	84
2 ^d	11b	45	12b	71
3 ^d	11c	50	12c	93
4 ^{c,f}	11d	50	12d	95
5 ^{b,g}	11e	55	12e	90

^aIsolated yield. ^bBetaine **1** (1.3 equiv) and Me₃SiCl (1.5 equiv) was added. ^cAqueous KF was used for hydrolysis step. ^dAqueous KOH was used for hydrolysis step. ^eGenerated in situ from imine and MeI. ^fMe₃SiCl (1.3 equiv) was added. ^gDichloroethane was used as solvent.

reaction can be visually monitored either by dissolution of poorly soluble reagent **1** or by gas evolution, which allows precise selection of reaction time and temperature (45–55 °C depending on substrate). When *p*-chlorobenzaldehyde was treated with betaine **1** at 50 °C, only a product of Wittig reaction,⁹ the corresponding difluoroalkene, was observed in the reaction mixture (¹⁹F NMR control). However, when the reaction was performed in the presence of a stoichiometric amount of Me₃SiCl, the desired product **12d** was isolated after the hydrolysis step in excellent yield (entry 4). Of special note is that in the latter case difluoroalkene was not detected. The reaction can also be applied to unbranched aliphatic aldehydes, as exemplified by difluoromethylation of dihydrocinnamaldehyde, with the best yield achieved using dichloroethane as solvent (entry 5). Concerning the effect of chlorosilane, we believe that the interaction of phosphonium ylide **2** with aldehydes generates betaine **13** which is rapidly trapped by

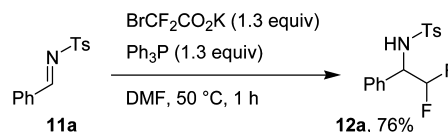
silylating reagent affording salt **14** (Scheme 4).¹⁸ Acetophenone was unreactive toward betaine **1** under standard conditions.

Scheme 4. Mechanism of Reaction of Betaine 1 with Aldehydes



Although betaine **1** is easily prepared,⁹ difluoromethylation can be performed simply starting from its precursors (Scheme 5). Indeed, heating of imine **11a** with potassium bromodifluoroacetate and triphenylphosphine with subsequent dephosphorylative treatment gave product **12a** in good yield.

Scheme 5. Difluoromethylation of Imine 11a



In summary, a practical method for nucleophilic difluoromethylation of various π -electrophiles using a shelf-stable and readily available phosphorus reagent is described. The key features of the reaction involve (a) its applicability to enolizable substrates due to moderate basicity of nucleophilic species and (b) a facile protodephosphorylation step, which proceeds even at room temperature.

■ ASSOCIATED CONTENT

S Supporting Information

Experimental procedures, compound characterization data, copies of NMR spectra for all compounds, and X-ray data for **5a** (CIF). This material is available free of charge via the Internet at <http://pubs.acs.org>.

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: adil25@mail.ru.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Russian Science Foundation (Project No. 14-13-00034).

■ REFERENCES

- (1) (a) Wang, J.; Sanchez-Roselló, M.; Aceña, J. L.; del Pozo, C.; Sorochinsky, A. E.; Fustero, S.; Soloshonok, V. A.; Liu, H. *Chem. Rev.* **2014**, *114*, 2432–2506. (b) Müller, K.; Faeh, C.; Diederich, F. *Science* **2007**, *317*, 1881–1886.
- (2) For selected reviews on trifluoromethylation reactions, see: (a) Prakash, G. K. S.; Yudin, A. K. *Chem. Rev.* **1997**, *97*, 757–786. (b) Liu, X.; Xu, C.; Wang, M.; Liu, Q. *Chem. Rev.* **2014**, DOI: 10.1021/cr400473a. (c) Dilman, A. D.; Levin, V. V. *Eur. J.*

- Org. Chem.* **2011**, 831–841. (d) Studer, A. *Angew. Chem., Int. Ed.* **2012**, 51, 8950–8958. (e) Shibata, N.; Matsnev, A.; Cahard, D. *Beilstein J. Org. Chem.* **2010**, 6, 65. (f) Ma, J.-A.; Cahard, D. *J. Fluorine Chem.* **2007**, 128, 975–996.
- (3) (a) Erickson, J. A.; McLoughlin, J. I. *J. Org. Chem.* **1995**, 60, 1626–1631. (b) Giornal, F.; Pazenok, S.; Rodefied, L.; Lui, N.; Vors, J.-P.; Leroux, F. R. *J. Fluorine Chem.* **2013**, 152, 2–11.
- (4) (a) Hu, J.; Zhang, W.; Wang, F. *Chem. Commun.* **2009**, 7465–7478. (b) Prakash, G. K. S.; Hu, J. *Acc. Chem. Res.* **2007**, 40, 921–930.
- (5) For radical difluoromethylation, see: Fujiwara, Y.; Dixon, J. A.; Rodriguez, R. A.; Baxter, R. D.; Dixon, D. D.; Collins, M. R.; Blackmond, D. G.; Baran, P. S. *J. Am. Chem. Soc.* **2012**, 134, 1494–1497.
- (6) For electrophilic difluoromethylation involving difluorocarbene, see: Ni, C.; Hu, J. *Synthesis* **2014**, 46, 842–863.
- (7) For transition-metal-catalyzed difluoromethylation, see: (a) Fujiwara, K.; Fujioka, Y.; Kobayashi, A.; Amii, H. *Org. Lett.* **2011**, 13, 5560–5563. (b) Fier, P. S.; Hartwig, J. F. *J. Am. Chem. Soc.* **2012**, 134, 5524–5527. (c) Prakash, G. K. S.; Ganesh, S. K.; Jones, J.-P.; Kulkarni, A.; Masood, K.; Swabeck, J. K.; Olah, G. A. *Angew. Chem., Int. Ed.* **2012**, 51, 12090–12094. (d) Jiang, X.-L.; Chen, Z.-H.; Xu, X.-H.; Qing, F.-L. *Org. Chem. Front* **2014**, 1, 774–776. (e) Matheis, C.; Jouvin, K.; Goossen, L. J. *Org. Lett.* **2014**, DOI: 10.1021/ol5030037. (f) Gu, Y.; Leng, X.; Shen, Q. *Nat. Commun.* **2014**, DOI: 10.1038/ncomms6405.
- (8) (a) Silicon reagent: Zhao, Y.; Huang, W.; Zheng, J.; Hu, J. *Org. Lett.* **2011**, 13, 5342–5345. (b) For sulfur reagents, see ref 4. (c) Phosphorus reagent: Beier, P.; Alexandrova, A. V.; Zibinsky, M.; Prakash, G. K. S. *Tetrahedron* **2008**, 64, 10977–10985. (d) Cadmium and zinc reagents: Burton, D. J.; Hartgraves, G. A. *J. Fluorine Chem.* **2007**, 128, 1198–1215.
- (9) Zheng, J.; Cai, J.; Lin, J.-H.; Guo, Y.; Xiao, J.-C. *Chem. Commun.* **2013**, 49, 7513–7515.
- (10) For applications of betaine **1**, see: (a) Qiao, Y.; Si, T.; Yang, M.-H.; Altman, R. A. *J. Org. Chem.* **2014**, 79, 7122–7131. (b) Zheng, J.; Lin, J.-H.; Cai, J.; Xiao, J.-C. *Chem.—Eur. J.* **2013**, 19, 15261–15266. (c) Deng, X.; Lin, J.; Zheng, J.; Xiao, J. *Chin. J. Chem.* **2014**, 32, 689–693.
- (11) In fact, ylide **2** has been used for a long time. It can be generated from different precursors, with betaine **1** being the most convenient one. For a recent overview, see: (a) Wang, F.; Li, L.; Ni, C.; Hu, J. *Beilstein J. Org. Chem.* **2014**, 10, 344–351. (b) For detailed discussion on ylide **2** and its earlier applications, see: Burton, D. J.; Yang, Z.-Y.; Qiu, W. *Chem. Rev.* **1996**, 96, 1641–1716.
- (12) Li, Q.; Lin, J.-H.; Deng, Z.-Y.; Zheng, J.; Cai, J.; Xiao, J.-C. *J. Fluorine Chem.* **2014**, 163, 38–41.
- (13) For our work on difluorocarbene-based nucleophilic reagents, see: (a) Kosobokov, M. D.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *Org. Lett.* **2014**, 16, 3784–3787. (b) Tsymbal, A. V.; Kosobokov, M. D.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *J. Org. Chem.* **2014**, 79, 7831–7835. (c) Levin, V. V.; Zemtsov, A. A.; Struchkova, M. I.; Dilman, A. D. *Org. Lett.* **2013**, 15, 917–919. (d) Zemtsov, A. A.; Kondratyev, N. S.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *J. Org. Chem.* **2014**, 79, 818–822. (e) Kosobokov, M. D.; Levin, V. V.; Zemtsov, A. A.; Struchkova, M. I.; Korlyukov, A. A.; Arkhipov, D. E.; Dilman, A. D. *Org. Lett.* **2014**, 16, 1438–1441. (f) Levin, V. V.; Dilman, A. D.; Struchkova, M. I. *J. Fluorine Chem.* **2014**, DOI: 10.1016/j.jfluchem.2014.08.021. (g) Smirnov, V. O.; Struchkova, M. I.; Arkhipov, D. E.; Korlyukov, A. A.; Dilman, A. D. *J. Org. Chem.* **2014**, DOI: 10.1021/jo5023537.
- (14) (a) Dilman, A. D.; Levin, V. V.; Belyakov, P. A.; Struchkova, M. I.; Tartakovsky, V. A. *Tetrahedron Lett.* **2008**, 49, 4352–4354. (b) Zemtsov, A. A.; Levin, V. V.; Dilman, A. D.; Struchkova, M. I.; Belyakov, P. A.; Tartakovsky, V. A. *Tetrahedron Lett.* **2009**, 50, 2998–3000. (c) Zemtsov, A. A.; Levin, V. V.; Dilman, A. D.; Struchkova, M. I.; Belyakov, P. A.; Tartakovsky, V. A.; Hu, J. *Eur. J. Org. Chem.* **2010**, 6779–6785. (d) Zemtsov, A. A.; Levin, V. V.; Dilman, A. D.; Struchkova, M. I.; Tartakovsky, V. A. *J. Fluorine Chem.* **2011**, 132, 378–381.
- (15) Lemek, T.; Mayr, H. *J. Org. Chem.* **2003**, 68, 6880–6886.
- (16) According to Mayr's scale, relative electrophilicities (*E* parameter) of Michael acceptors decrease in the following order for benzylidene derivatives: Meldrum's acid (−9.15) > malononitrile (−9.42) > malonic ester (−20.55). See: Kaumanns, O.; Lucius, R.; Mayr, H. *Chem.—Eur. J.* **2008**, 14, 9675–9682.
- (17) Dilman, A. D.; Arkhipov, D. E.; Levin, V. V.; Belyakov, P. A.; Korlyukov, A. A.; Struchkova, M. I.; Tartakovsky, V. A. *J. Org. Chem.* **2008**, 73, 5643–5646.
- (18) We detected salt **14** by ^{19}F NMR of reaction mixture. A similar salt was previously observed in reaction of ylide **2** with an aldehyde and Me_3SiBr ; see ref 11a.